Science and Medicine in Football

ISSN: 2473-3938 (Print) 2473-4446 (Online) Journal homepage: www.tandfonline.com/journals/rsmf20

How to mitigate injury risk after return-to-play in elite football? Expert judgement meets data analytics

Anne Hecksteden , Julian Berger , Raffaele Mazzolari , Guangze Zhang , Tim Meyer, Michel Brink , Koen Lemmink , Josef Fritz , Niklas Keller & Thomas Hauser

To cite this article: Anne Hecksteden , Julian Berger , Raffaele Mazzolari , Guangze Zhang , Tim Meyer, Michel Brink , Koen Lemmink , Josef Fritz , Niklas Keller & Thomas Hauser (09 Mar 2026): How to mitigate injury risk after return-to-play in elite football? Expert judgement meets data analytics, Science and Medicine in Football, DOI: 10.1080/24733938.2026.2642654

To link to this article: https://doi.org/10.1080/24733938.2026.2642654

© 2026 The Author(s). Published by Informa UK Limited, trading as Taylor & Francis Group.

View supplementary material

Published online: 09 Mar 2026.

Submit your article to this journal

View related articles

View Crossmark data

Full Terms & Conditions of access and use can be found at
https://www.tandfonline.com/action/journalInformation?journalCode=rsmf20

SCIENCE AND MEDICINE IN FOOTBALL
https://doi.org/10.1080/24733938.2026.2642654

Routledge
Taylor & Francis Group

OPEN ACCESS Check for updates

How to mitigate injury risk after return-to-play in elite football? Expert judgement meets data analytics

Anne Hecksteden[a,b], Julian Berger[c], Raffaele Mazzolari[a], Guangze Zhang[d], Tim Meyer[d], Michel Brink[e], Koen Lemmink[e], Josef Fritz[f], Niklas Keller[g*] and Thomas Hauser[h,i*]

[a]Institute of Sport Science, University of Innsbruck, Innsbruck, Austria; [b]Institute of Physiology, Medical University of Innsbruck, Innsbruck, Austria; [c]Center for Adaptive Rationality, Max Planck Institute for Human Development, Berlin, Germany; [d]Institute of Sports and Preventive Medicine, Saarland University, Saarbrücken, Germany; [e]Department of Human Movement Sciences, University of Groningen, University Medical Center Groningen (UMCG), Groningen, The Netherlands; [f]Institute of Clinical Epidemiology, Public Health, Health Economics, Medical Statistics and Informatics, Medical University of Innsbruck, Innsbruck, Austria; [g]Harding Centre for Risk Literacy, Faculty of Health Science, University of Potsdam, Potsdam, Germany; [h]Faculty of Applied Sport Sciences & Personality, Business and Law School, Berlin, Germany; [i]Fédération Internationale de Football Association, High Performance Department, Zürich, Switzerland

ABSTRACT

In elite football, injury risk is considerable, especially in the weeks after return-to-play (RTP). Consequently, optimal rehabilitation and well-informed decisions regarding the timepoint of RTP are crucial. Arguably, experiential knowledge as well as data-driven analyses may provide meaningful insights. This work leverages their synergies to identify and scrutinize promising options for reducing injury risk. A mixed-methods approach is implemented as sequential problem-solving pipeline. Semi-structured interviews were conducted with 14 experienced practitioners in elite European football. Areas of overlap in the interview data were identified by thematic analysis. Subsequently, media-based injury data was used to scrutinize expert judgement. In total 4 themes were identified concerning 1) time between index injury and RTP (rehab time), 2) rehabilitation strategy, 3) psychosocial factors, and 4) team structure. Most prominently, it was suspected that injury risk post-RTP would be lower with longer rehab times. Data analytics confirmed lower injury risk after longer rehab time (for the respective index injury diagnosis) if expressed per calendar day but not if expressed per unit of football exposure post-RTP. Hazard ratios for pooled episodes were 0.886 (0.834–0.937) for calendar time in days and 1.046 (1.005 – 1.087) for football exposure in weighted hours. Taken together, our results indicate options for improvement which are simple (at least logistically) and largely consistent with the literature. However, quantitative results don't provide direct support to the experts judgement that delaying RTP would reduce injury risk. Importantly, these findings are likely context specific and pertain only to elite players who sustained a subsequent injury.

ARTICLE HISTORY
Accepted 22 January 2026

KEYWORDS
Mixed-methods; expert elicitation; causal inference; soccer

Introduction

Exercise-related injuries are an issue in many sports including football. In professional players, about two time-loss injuries may be expected per season (Ekstrand et al. 2011, 2021) with recurrency rates for common injuries between ~10 and 25% (Arnason et al. 2004; Hagglund et al. 2016; Ekstrand et al. 2021). Considering the negative consequences of injuries and the ensuing layoffs for individual players and their teams (Hagglund et al. 2013; Ekstrand et al. 2021), understanding, assessing, and mitigating injury risk is of major importance.

Over the last decades, massive research efforts have enabled progress from identifying factors associated with an increased risk of injury ('risk factors') to the development of complex causal concepts (Bittencourt et al. 2016), high-dimensional, machine learning (ML) based risk assessment approaches (Ruddy et al. 2019; Hecksteden et al. 2023), and targeted preventive measures (Hagglund et al. 2016; Obertinca et al. 2024). Clubs and teams, too, have dedicated impressive resources to implementing preventive measures (Lutter et al. 2022) as well as to collecting and analysing data related to injury risk (Bittencourt et al. 2016). It is therefore striking that during this time period only a slight decrease in injury incidence

CONTACT Anne Hecksteden anne.hecksteden@uibk.ac.at Department of Sport Science, Universität Innsbruck, Chair of Sports Medicine, Fürstenweg 176, Innsbruck 6020, Austria

*Authors contributed equally.

Supplemental data for this article can be accessed online at https://doi.org/10.1080/24733938.2026.2642654.

© 2026 The Author(s). Published by Informa UK Limited, trading as Taylor & Francis Group.
This is an Open Access article distributed under the terms of the Creative Commons Attribution License (http://creativecommons.org/licenses/by/4.0/), which permits unrestricted use, distribution, and reproduction in any medium, provided the original work is properly cited. The terms on which this article has been published allow the posting of the Accepted Manuscript in a repository by the author(s) or with their consent.

(Ekstrand et al. 2021) and recurrency rates (Hagglund et al. 2016) has been observed for professional 2Plausible reasons for the lagging translation of scientific progress into practical benefit are diverse. Clearly, tailored implementation (Benjaminse and Verhagen 2021; Edouard et al. 2022; Tabben et al. 2023) and good communication (Ekstrand et al. 2019) are crucial for fostering adoption and adherence which in turn are prerequisites for effectiveness. Moreover, the preventive effects may be partially outweighed by increasing physical demands of the game (Barnes et al. 2014). From a methodological perspective, sports injury research is characterized by limited case numbers, a multitude of influencing factors, and a probabilistic outcome (with chance playing a role in injury occurrence). This combination of challenges, while not unique to sports injury research in high-performance athletes, can limit the predictive value of typical scientific studies. Therefore, complementary approaches could be beneficial to address questions related to sport injury research (Greenland et al. 2016; Hecksteden et al. 2022; Hecksteden A, Keller N, et al. 2023).

Finally, it is important to note that risk mitigation and risk estimation are two distinct aspects of successful (injury) risk management. Risk mitigation requires evidence of efficacy, that is: causal knowledge (e.g., 'If we postpone RTP by two weeks – will injury risk be lower than if he returns today?'). By contrast, practically helpful estimation of injury risk for a given player and timeframe (that is: for one specific set of determinants) critically depends on the accuracy of risk estimates in a practical, prospective setting ('If this player returns to full availability today, what is the probability that he will get injured in the upcoming four weeks?') (Hecksteden A, Keller N, et al. 2023; Hecksteden et al. 2023). While this distinction applies to injury risk in general, it is particularly salient in the context of return-to-play (RTP) after a previous injury. On the one hand, identifying a reasonable timepoint for RTP (RTP decision making) requires at least an implicit risk estimate for the individual case (Shrier 2015; Yung et al. 2022) – which will generally be above baseline level (that is: Above the injury risk of the individual player within the competitive season but outside the post-RTP timeframe) (Orchard and Best 2002; Hagglund et al. 2007; Gajhede-Knudsen et al. 2013; Wangensteen et al. 2016). This estimate can then be evaluated by the decision maker against what is deemed acceptable in a specific situation (Shrier 2015). On the other hand, the rehabilitation process aims to restore loading and performance capacity, thereby minimizing the risk of a recurrence or other subsequent injury after RTP. From this perspective it is important to identify existing best practices as well as options for improvement: 'Screws' which – if turned – will improve rehab outcomes.

Note that in theory a comprehensive, fully specified causal model could serve both tasks by providing accurate risk estimates as well as valid projections for the effect of interventions (although predictive performance will remain limited by the element of chance in injury occurrence). However, as mentioned above, this is unrealistic under the framework conditions of sport injury research (Hecksteden et al. 2022, 2023; Hecksteden A, Keller N, et al. 2023).

In search of ways to deal with these epistemic constraints, it is important to note that scientific investigation is not the only way of addressing gaps in knowledge (e.g., concerning injury risk assessment and mitigation). In fact, experiential knowledge and human judgement are also recognized ways of supporting good decisions in practice. Specifically in the field of sport, scientific evidence (e.g., from experimental trials, epidemiological studies, or qualitative research) and practitioners' experiential knowledge may have synergistical strengths and weaknesses (Hecksteden et al. 2022; Hecksteden A, Keller N, et al. 2023). Moreover, while the two ways of knowledge generation are still often perceived as 'either-or' alternatives, today a variety of methods for their integration are available (Hecksteden et al. 2022; Hecksteden A, Keller N, et al. 2023). Examples range from data-driven verification of hypotheses derived from expert opinion over feature engineering for machine learning (Rudin 2019; Hecksteden et al. 2023) and the elicitation of priors for Bayesian analyses (van de Schoot et al. 2021), to intricate multi-stage human-computer interaction in an interpretable machine learning context (Rudin et al. 2021; Rudin 2019).

Based on these considerations, the aim of this study is to leverage synergies between different ways of knowledge generation in the context of rehabilitation and RTP in elite football. Experiential knowledge regarding options for reducing injury risk after RTP will be characterized based on interviews with stakeholders from the highest level of play. Subsequently, targeted data analytics will be used to scrutinize expert judgment. This specifically concerned the expectation that injury risk would be lower if RTP would be delayed. Finally, qualitative as well as quantitative results will be discussed in the background of previous literature.

Methods

General approach

This work follows a mixed-methods approach (Timans et al. 2019), implemented as a sequential problem-

solving pipeline. Its core consists of semi-structured interviews with experienced practitioners in elite football. In agreement with previous work (Hecksteden et al. 2023), RTP was defined as full availability of the player for training and matches. Thematic analysis guided by the six-phase model by Braun and Clarke (Braun et al. 2016) was used for identifying (explicit as well as latent) areas of overlap between expert viewpoints. Targeted analyses of injury data from professional football as well as published scientific work (e.g., regarding epidemiological indicators, biological timelines, and causal concepts) were used to reflect, verify, and contextualize the interview results, aiming at identifying areas of concordance and understanding discrepancies. The study was approved by the institutional review board of the Institute of Sport Science at the University of Innsbruck (approval number 57/2023 and amendment). Conduct and reporting of the qualitative work were informed by current guidelines (O'Brien et al. 2014). The SRQR checklist is provided as a supplement.

In the quantitative part of this work, the exposure of interest, for which the effect on injury risk after RTP is investigated, is rehab time (calendar days elapsed between the index injury and RTP, relative to the index injury diagnosis). This reflects the analytical aim of examining theme 1, which suggests that actively extending rehab time reduces injury risk post-RTP. Note that focusing on remaining deficits would be more appropriate when seeking biological explanation. The occurrence of a subsequent injury after RTP is evaluated using a time-to-event approach with two alternative time axes: The conventional time axis in time-to-event analysis is chronological (here: calendar time in days). However, for injuries in football the actual time-at-risk is during training and match play (football exposure) only. Aiming at an indicator that quantifies the global amount of opportunity to get injured, in this work football exposure post-RTP is calculated as the sum of match and training times, weighted for the difference in injury incidence during training and match play, respectively. Details are provided in the section 'Calculated indicators' below and in the supplemental code file.

Note that the original interviews concerned the clinical, risk mitigation perspective of improving the rehabilitation process and minimizing injury risk as well as the forecasting perspective of estimating the risk of subsequent injury for a given player and context. To ensure readability, results regarding the forecasting perspective will be reported in another manuscript.

Interviews

Interviewees

The interviews elicited expert opinions concerning options for optimizing rehabilitation and post-RTP player management with the goal to minimize post-RTP injury risk. All professions involved in the management and rehabilitation of injuries including RTP decision making were eligible. Care was taken to include a diverse set of qualifications and roles covering the whole rehabilitation process (orthopaedic surgeons, physiotherapists, rehab, strength-and-conditioning coaches, and embedded sport scientists). Individuals were required to be involved in top-level European football (national first leagues or national teams including associated medical facilities) for at least 5 years prior to the interviews.

Interview setting

Data were collected between March and November 2023. A semi-structured interview format was used. The framing was a hypothetical 'forecasting tournament' in which the participants are required to provide probability estimates for a subsequent injury within 28 days at the day of RTP of the respective player. The timeframe has been informed by preparatory work on the time course of injury risk after RTP in football (Zhang et al. 2025). Interviews were generally conducted in person, with three exceptions which were conducted via videoconference. With one exception, interviews were conducted in the native language of the interviewee (German or Dutch). To avoid restraint on the side of the participants, the interviews were not recorded but documented in a written protocol. The target duration of the interviews was 60 minutes. All interviewees provided informed consent prior to participation. The interview material is provided in the supplements.

Thematic analysis

Thematic analysis was used to identify and interpret areas of overlap in the viewpoints conferred by the expert practitioners (Braun et al. 2016). Thematic analysis is a principled, yet flexible and pragmatic analytical approach which allows to identify patterns of meaning in a qualitative dataset (e.g., interviewee responses) which fits well with the framework conditions in elite sports (Hecksteden et al. 2022; Hecksteden A, Keller N, et al. 2023). Explicit as well as implicit themes and subthemes were considered. Care was taken to maintain an inductive perspective throughout, focusing on the interview data while limiting application of preexisting concepts and categories.

We followed the six-phase model of thematic analysis described by Braun and Clarke (Braun et al. 2016). Authors AH, GZ, and MB familiarized themselves with all interview protocols. AH coded the interview protocols and developed a preliminary set of candidate themes and subthemes which was discussed and revised by AH, GZ, and MB until consensus was reached. As a final step, themes and subthemes were characterized with concise titles and definitions by AH.

Researcher characteristics and reflexivity

The researchers involved in data collection and analysis (AH, MB, GZ) have an academic background in sports medicine and sport science with a research focus in professional football. AH and MB have faculty positions while GZ has been a PhD candidate at the time of conducting study. The work of GZ was funded by a scholarship from the German Football Association (DFB). No other external funding was used. There were no conflicts of interest. While interviewees were recruited via the respective networks of AH, TM and MB, there was no collaboration (previous or ongoing), dependency or personal relation (beyond acquaintance). Care was taken by the core research team to maintain reflexivity throughout the study by writing down personal assumptions and expectations prior to the start of data collection, keeping a reflexive diary during analysis, and discussing potential impact of both during analysis and interpretation.

Data analytics – data-driven scrutiny of interview results

Some of the prioritized themes identified in the expert interviews had not been specifically addressed in the literature so far. We therefore used targeted analyses of publicly available injury data to verify the expert judgement as far as possible. Data analysis was conducted in R (version 4.4.0). A concise outline is provided below. Densely commented code and raw data are provided in the supplements.

The effect of rehab time on injury risk – do they really return too early?

Theme 1 posits a causal relationship between rehab time and injury risk. Specifically, it refers to a positive effect of increasing rehab time by consciously delaying RTP – either generally or only for very short rehab times. While similar suspicions have been repeatedly brought forward in the literature based on indirect evidence (Beischer et al. 2020; Bengtsson et al. 2020; Della Villa et al. 2021; Ekstrand et al. 2021; Gajhede-Knudsen et al. 2013; Orchard and Best 2002; Stares et al. 2018; Wangensteen et al. 2016), to the best of our knowledge a potential causal relationship has not been explicitly investigated so far. To address this gap in the available evidence and specifically scrutinize the expert judgement, we propose an analysis based on the assumptions illustrated by the causal DAG (directed acyclic graph) depicted in Figure 1. Note that there are two potential

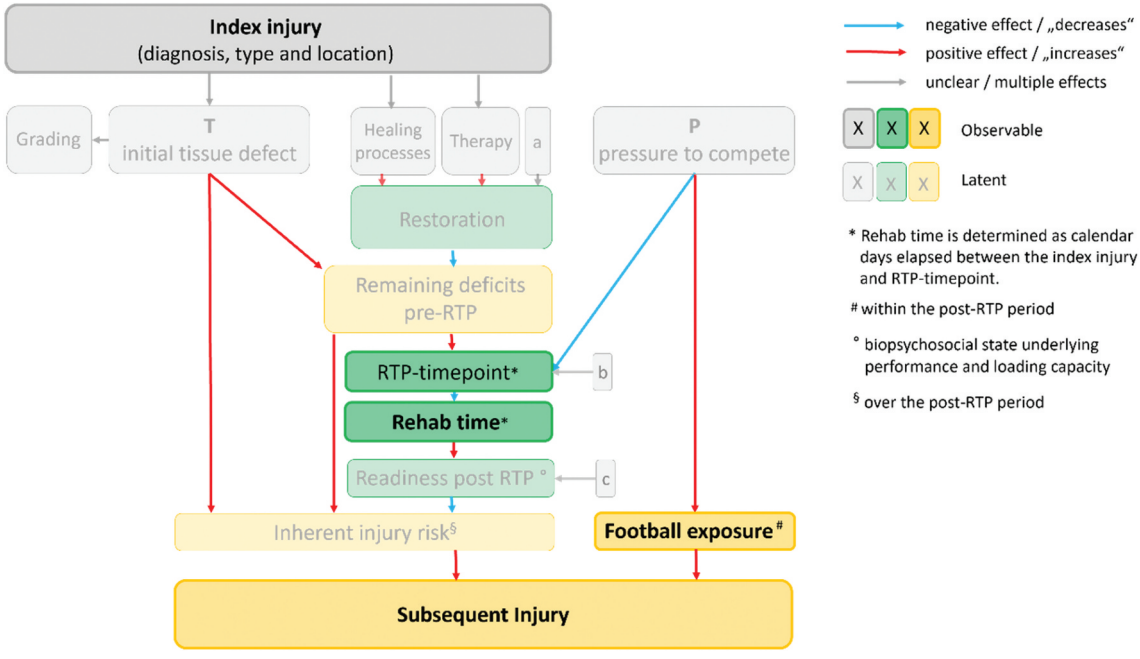

Figure 1. Causal DAG. a and b represent potential unobserved influencing factors. C represents influencing factors of readiness post-RTP other than remaining deficits following from the index injury.

confounders of the association between rehab time and injury: 'pressure to compete' (P) and 'initial tissue defect' (T, the within-diagnosis grade of the index injury). While both confounders are not directly observable, P has an observable mediator ('football exposure' post-RTP; E) that can be used as a surrogate. Note further, that T would plausibly cause a positive association between rehab time and injury risk (as larger initial tissue defect would lead to episodes with long rehab time and higher risk, e.g., due to more defect-healing or deconditioning – and vice versa). Therefore, T cannot explain a negative association between rehab time and injury risk. This allows us to focus on P as a potential alternative explanation for any observed negative association between rehab time and injury risk. Importantly, this is a conservative approach in the sense that the causal effect of rehab time may be plausibly under- but not overestimated.

Injury data

Time-loss injuries and match exposures were collected from online sources using a custom PHP script. All time loss injuries concerning the musculoskeletal system were eligible as index and subsequent injuries. The main source was www.transfermarkt.com. Specifically, players competing in the European top leagues (first leagues in Austria, France, Germany, Italy, the Netherlands, Spain, and United Kingdom) during the season 2023/24 were identified and followed back in time until their debut in professional football. Subsequently, index injury-RTP-subsequent injury episodes were constructed. Note that, when considering the injuries of a particular player in chronological order, the subsequent injury of episode n will also be the index injury of episode n + 1.

Considering the limitations of media-based injury data specifically regarding medical details (Hoenig et al. 2022), we focused on pooled data (all index injuries) and clearly identifiable, typical football injuries ('big 3': ankle sprain, anterior cruciate ligament rupture, thigh muscle injury). The heterogeneity of these practically relevant index injury categories is deliberate and leveraged to scrutinize the robustness of results.

In accordance with the framing of the interviews, the duration of post-RTP follow-up was 28-days. This specific time frame was based on preparatory work indicating that the excess injury risk after RTP mostly decays within the first 4 weeks (Zhang et al. 2025).

Calculated indicators

Rehab time. Rehab time was calculated as the time loss between index injury and RTP. When episodes are analysed by index injury diagnosis, rehab time is expressed in fortnights (not days) to allow for meaningful hazard ratios. When episodes are pooled across index injury diagnoses, differences in the distribution of rehab time must be considered (e.g., a time loss of 3 months would be very long for an ankle sprain but very brief for an ACL tear). A robust scaling approach was used to make rehab time comparable between diagnoses. Specifically, scaled rehab times were calculated by taking the difference between the observed time loss and the within-diagnosis median time loss, and dividing it by the within-diagnosis interquartile range (11). Scaled rehab time thereby quantifies how long or short rehab time had been for the respective index injury diagnosis. Note that including the grouping variable in the model would be an alternative approach. We choose robust scaling based on its transparency as well as in consideration of the large differences in case numbers and rehab times between diagnoses. For clarity, the alternative approach is demonstrated in the supplementary code file.

Football exposure post-RTP. Football exposure within the 28-day follow-up was estimated based on observed match exposures. While match exposures were collected on an individual basis and merged with the injury data, training exposure was estimated based on typical values in European pro soccer as reported in the literature (225 hours per season based on Ekstrand et al. 2011, 2021). To allow meaningful integration of match and training exposures (ME and TE, respectively) into an indicator for the 'magnitude of opportunity to get injured', training times were weighted according to the lower risk of injury during training as compared to match play (factor 1/8 based on Ekstrand et al. 2021; Aus der Funten et al. 2023). In short, cumulative football exposure during follow-up (CumEx) was calculated as: CumEx = ME [minutes] + (TE [minutes]/8).

Data analysis

Rehab time and injury risk post-RTP: time-to-event analysis. Aiming at transparency while also considering the requirements and limitations of potential analytical methods, a time-to-event approach (Cox proportional hazards model) was used. Rehab time was used as continuous predictor. Player ID was included as a random effect to account for the nesting of episodes within players. The main model uses weighted hours of football exposure post-RTP (CumEx) as time axis, thereby modelling instantaneous risk (hazard) on that scale. Note that this model reflects the causal assumptions illustrated in the causal DAG (Figure 1) as well as recommendations concerning time-at-risk in collecting and analysing sports injury data (Bahr et al. 2020). A secondary

comparator model uses the habitual chronological time axis (calendar time in days).

We are cognizant that this approach, while based on the causal DAG in Figure 1, does not implement 'causal inference', as it does not directly provide an unbiased estimate of the causal effect of rehab time on injury risk over the 28-day follow-up. However, comparison between the two models – as well as to the results of previous work based on chronological timeframes – allows for scrutinizing whether a negative association between rehab time and injury risk might be plausibly accounted for by football exposure post-RTP.

Survival curves with the respective time axes are used for illustration. Please note that this requires categorizing rehab time. Average rehab time was defined for each index injury diagnosis as 0.5 interquartile ranges above and below the median. With short being below average and long above.

The level of significance was set at $\alpha < 0.05$. Hazard ratios are reported with 95% confidence intervals. Reproducible details are provided in the supplemental code file.

Potential threshold effect for rehab time: ROC curves. Subtheme 1.2 differs from subtheme 1.1 in that it looks specifically on very short rehab times at the lower border of what is possible for a specific index injury diagnosis. It expresses the supposition that the proportion of players with a subsequent injury in the first 28 days after RTP increases abruptly when rehab time drops below a critical threshold (e.g., set by the time course of key biological healing processes). A systematic way of searching for a threshold in a continuous predictor for a binary outcome is the receiver operating characteristic (ROC) curve (Katsikopoulos et al. 2021). A localized bend in the ROC curve would indicate a potential threshold effect. Note that here ROC curves are not used to assess classificatory performance by looking at the area under the curve (AUC), which is expectedly low.

Results

Interview results

Interviewees

In total, 14 interviews could be analysed (2 orthopaedic surgeons, 5 physiotherapists, 7 rehab and/or strength-and-conditioning (S&C) coaches/embedded sport scientists). Note that except for the orthopaedic surgeons, the distinction between qualifications and roles was not exclusive. Several interviewees had a double or even triple qualification (physiotherapist, UEFA A coaching licence, master's degree in sport science) and some of them assumed two roles (physiotherapy and rehab coach or rehab and S&C coach). In this case they were classified according to the focus of their current role. At the time of the interviews, participants were active in Germany (7), Austria (2), the Netherlands (4) and the United Kingdom (1).

Interview results

Table 1 summarizes the four themes and six subthemes in approximate order of importance and consistency in the interview data. Most notably, experts expect a decrease in injury risk if RTP would be delayed.

Data analytics

Figure 2 illustrates sample size for the quantitative analyses.

Subgroup analyses are secondary. Note that the subsequent injury in episode n of a specific player is also the index injury of episode n + 1. Incomplete episodes arise from the last (or only) injury of a player and were excluded.

Rehab time and injury risk after RTP

Table 2 summarizes the results of hierarchical Cox regression with the two alternative time axes and rehab time as predictor variable. For episodes pooled across index injury diagnoses (grey background in Table 2), injury rate was lower after longer rehab time (for the respective index injury diagnosis) only if expressed per calendar day but not if it is expressed per hour of football exposure post-RTP. Note that the confidence interval in the latter case ($HR_{football}$) is even entirely above 1. While this finding is unexpected, it may be speculated that short rehab time becomes an indicator of low initial tissue defect (or highly effective rehab) once football exposure post-RTP is accounted for.

Regarding subgroup analyses, the hazard ratio for a time axis in calendar days (HR_{days}) is numerically below 1 in all cases. Moreover, $HR_{football}$ is numerically larger than HF_{days} in all cases except recurrent ACL tears. However, statistical significance is not reached when analyses are performed by subgroup. Note the variation in the numbers of episodes and observed events when comparing results between subgroups.

Survival curves for the main dataset are visualized in Figure 3. As rehab time is the exposure of interest, the focus is on differences in survival between rehab

Table 1. Theme overview.

Theme 1: "They return too early."
When the risk of a subsequent injury is taken as the criterion, players generally return to early.
Subtheme 1.1: "More rehab time would considerably reduce injury risk."
A gradual decrease of injury risk with longer rehab even for typical ("reasonable") rehab times. The suspected mechanisms included incomplete tissue maturation ("Ligamentation takes two years.") as well as more time for (athletic and football specific) conditioning and ensuring psychological readiness.
Subtheme 1.2: "Biological constraints cannot be overcome."
An abrupt, massive increase in injury risk with "unreasonably" low RTP times that conflict with the timeline of key healing processes that restore adequate loading capacity of the injured structure. Conversely, a reduction in risk with longer rehab would be expected only for rehab times at the low end of the spectrum.
Subtheme 1.3: "Come back stronger."
The potential for using rehab time to address longstanding deficits and eventually reduce injury risk below pre-injury levels (and/or increase performance). While this subtheme connects to and extrapolates subtheme 1.1, there was lesser emphasis and consensus on this point. Specifically, opinions differed whether this is restricted to rare cases with clearly identifiable pre-injury deficits as well as about the practicability of the required timelines.
Theme 2: "Mission command - Consistent strategy & adaptive, outcome-oriented Execution"
Options for improvement of the rehab process revolved around prioritizing a "humble", approach based on a consistent strategy, proven measures, and a "baby steps" approach with tight feedback loops.
Subtheme 2.1: Smoothing-out gaps and leaps
Losses in informal information and therapeutic consistency at transition points between rehab phases or therapists was identified as a key option for improvement which could be easily addressed. Similarly, an approach to progression based on small steps with sincere feedback loops depending on the players response was favored over more clearcut transitions between rehab phases. Importantly, this did not question the value of objective milestones at larger intervals which ensure overall structure, minimal requirements and communication. Importantly, this consideration included post-RTP player management.
Subtheme 2.2: Small steps - but no "cowardly rehab"!
Full loading should be obtained during the individually guided rehab process - even if this may entail the risk of negative responses/injuries during rehab (as opposed to after RTP).
Subtheme 2.3: Better safe than cool.
During the rehab process, a consistent, dependable package of measures and the quality of their execution are key. In other words: It is important to do what is proven and mastered – not the hot new stuff. Along the same line, skipping steps during the rehab process is strongly discouraged.
Theme 3: Squeezes and focus – Psychosocial factors deserve more attention.
Internal as well as external pressure (e.g., due to career situation or the head coach) and psychosocial stressors may lead to pushing through minor complaints before and after RTP. Also, emotional recovery and the restored ability to focus on the game are to be considered. When aiming to minimize injury risk, a psychological "safety bubble" should be provided for the player during rehab and the post-RTP period.
Theme 4: "Small can be beautiful." – The downsides of large rehabilitation teams
The increasingly large medical teams also have downsides including unclear responsibilities, extensive communication, and information loss. Specifically, the need to prioritize the suitability for interpersonal communication as opposed to the ecological validity of findings (e.g., GPS data over the "coaches eye") may plausibly have negative consequences.

time categories on each of the two alternative time axes. When comparing the two panels, it is important to note that the endpoint of a Kaplan-Meier survival curve is not only dependent on the proportion of observed events. Rather, for the same proportion of observed events, early censoring and late events lead to a lower endpoint of curve compared to early events (because the 'steps' in the curve are higher with a lower number of players at risk at the timepoint of the respective event). Please also compare the rendered code in the supplements.

Further visualizations are provided in the supplementary code file. While they are generally consistent with the proposed causal model, it seems particularly noteworthy that high football exposures post-RTP occur virtually only after early return and that this is a high-risk constellation (cp. last figure in the supplementary code file). Note that this finding remains preliminary, as the adjustment of football exposure for the timepoint of an eventual subsequent injury exaggerates football exposure for cases with a subsequent injury very early after RTP, which would at least partially explain the differences.

Is there a threshold of rehab time below which injury risk increases abruptly?

A potential threshold effect for rehab time (as posited by theme 1.2 'Biological constraints cannot be overcome.') would be indicated by a localized bend in the ROC curve. Note that this differs from assessing classificatory performance based on area under the ROC curve (AUC), which is expectedly low. Our results, illustrated in Figure 4, do not suggest a threshold effect for rehab time.

Discussion

Would injury risk after RTP be lower with longer rehab times?

We identified options for improving the rehabilitation process and reduce injury risk after RTP in elite football. One contribution with the potential for considerable scientific and practical relevance is a first scrutiny of a potential causal relation between rehab time and injury risk post-RTP. The main result from the interviews was the large consensus between expert practitioners that injury risk after RTP would be lower with more rehab time – either

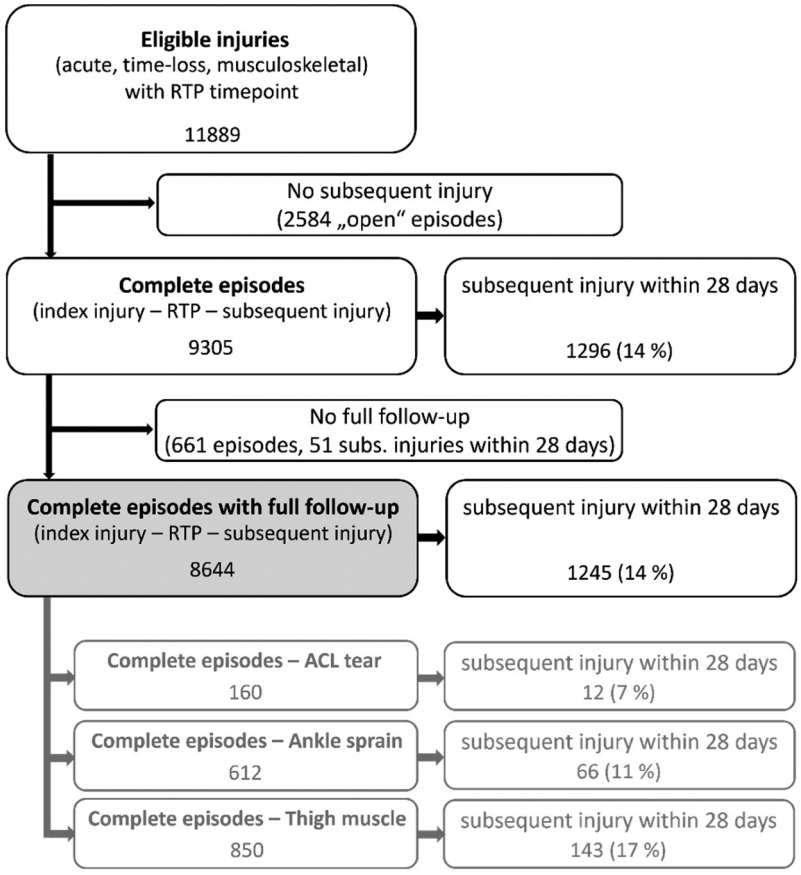

Figure 2. Sample flowchart. Grey background: dataset for the main analysis.

Table 2. Results of hierarchical Cox models with different time axes.

	n	HR_{days}	$HR_{football}$
All subsequent injuries			
All index injuries		Time loss scaled within diagnosis	
	8644 (1245)	**0,886 (0,834 – 0,937)**	**1,046 (1,005 – 1,087)**
By index injury diagnosis		Time loss per fortnight	
ACL tear, complete	160 [12]	0,940 (0,821 – 1,059)	0,941 (0,829 – 1,053)
Ankle sprain	612 [66]	0,907 (0, 798 – 1,016)	0,978 (0,874 – 1,082)
Thigh muscle injury	850 [143]	0,980 (0,903 – 1,057)	1,072 (1,000 – 1,145)
Subsequent injury: Same diagnosis as index injury			
All index injuries		Time loss scaled within diagnosis	
	2812 [535]	**0,800 (0,707 – 0,893)**	0,974 (0,893 – 1,055)
By index injury diagnosis		Time loss per fortnight	
ACL tear, complete	19 [5]	0,844 (0,621 – 1,067)	0,714 (0,331 – 1,097)
Ankle sprain	93 [23]	0,909 (0,727 – 1,091)	0,936 (0,760 – 1,113)
Thigh muscle injury	185 [48]	0,873 (0,711 – 1,035)	0,999 (0,829 – 1,169)

Grey background: Main analysis (pooled episodes, all subsequent injuries)
n: Number of episodes [number of observed events]
HR_{days}: Hazard ratio, time axis in calendar days since return-to-play
$HR_{football}$: Hazard ratio, time axis in units of football exposure since return-to-play
Ranges in brackets are 95% confidence intervals
bold: $p < 0{,}05$.

generally, or only but massively for very short rehab times that drop below a biological threshold. In the scientific literature, a negative correlation between rehab time and the risk of early recurrence has been reported for a number of typical football injuries including complete ACL tears (Beischer et al. 2020; Della Villa et al. 2021), Achilles tendinopathies (Gajhede-Knudsen et al. 2013), and muscle injuries (Stares et al. 2018). From a physiological perspective, a strong case can be made that typical rehab times conflict with the timeline of healing processes leading to

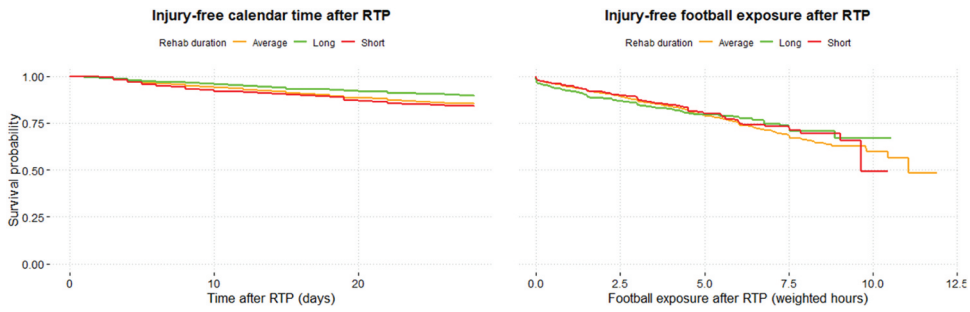

Figure 3. Survival curves with the two alternative time axes (main dataset). Average rehab time was defined for each index injury diagnosis as 0.5 interquartile ranges above and below the median. With short being below average and long above.

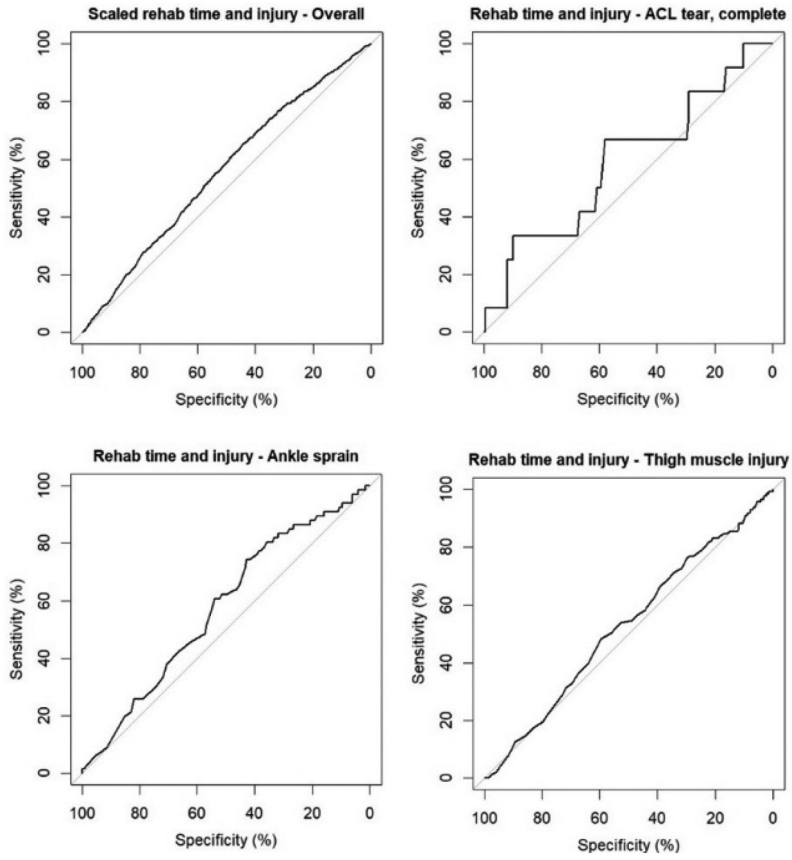

Figure 4. ROC curves.

suboptimal loading capacity of the injured structure at the timepoint of RTP (Orchard and Best 2002). The typical occurrence of reinjuries early after RTP also points to an incomplete (and functionally insufficient) local healing (Orchard and Best 2002; Wangensteen et al. 2016). While in principle this is in favor of a causal relationship between rehab time and injury risk (and therefore of reduced risk with prolonged rehab), it must be kept in mind that this evidence is indirect.

To the best of our knowledge, no investigations have been published that explicitly investigate the causal effect of rehab time and/or base their analyses on explicit causal assumptions. We therefore probed whether the negative association between rehab time and injury risk which we and others observed (Gajhede-Knudsen et al. 2013; Stares et al. 2018; Beischer et al. 2020; Bengtsson et al. 2020; Della Villa et al. 2021; Ekstrand et al. 2021) can be explained by confounders, specifically football exposure post-RTP as a mediator of pressure to compete (cp. Figure 1). Our results confirm hazard ratios below 1 – but only when time-at-risk is quantified in chronological units (calendar days) but not when

quantified in units of football exposure (weighted hours). This finding suggests that the higher risk after early return is at least in part accounted for by higher football exposure post-RTP. Our quantitative results do not provide evidence of a causal relationship.

Important aspects to consider when drawing conclusions from this result

Rehab time, load management, and survivorship bias. An important aspect related to associations between rehab time and injury risk are selection effects during rehabilitation. Plausibly, more rehab time and accumulated load in this period not only favor healing and conditioning but also increase the probability of sustaining an injury before (and not after) the next milestone. While this is similarly detrimental for the individual player and his team, the 'weeding out' of frail cases during rehabilitation may plausibly contribute to lower injury risk after RTP with longer rehab and/or higher accumulated load during rehabilitation. Interestingly, selection effects during rehabilitation effects are rarely discussed in the pertinent literature, e.g. Bengtsson et al. (2020). As potential alternative explanations for the observed association. Note that this aspect is also closely connected to subtheme 2.2 (Small steps – but no 'cowardly rehab'!) discussed further below.

Context dependence. When interpreting the above result, it is important to keep in mind that the relationship of rehab time and injury risk is investigated in a high-performance environment. On the one hand this means a highly selected and highly trained population of players. On the other hand, it means highly qualified and experienced medical and coaching staff. This plausibly limits the variation in RTP decision making, resulting in few 'unreasonably' short (or long) rehab times which could drive an association with injury risk. This is especially relevant for subtheme 1.2. ('Biological constraints cannot be overcome.') In fact, recent work points to an increasing appreciation for the importance of adequate rehab time for successful RTP (Ekstrand et al. 2021). However, it is also pointed out that for certain diagnoses including ACL and syndesmosis injuries more and longer rehab may still be desirable (Ekstrand et al. 2021). In brief, it is highly likely that 'unreasonably' short rehab times would increase injury risk. It is less clear how often and to what extent such cases occur in current elite football.

What about bias from confounders. The second confounder in the causal DAG 'initial tissue defect' (T, the within-diagnosis grade of the index injury, whether knowable or not) has not included in our model because it had not been measured. It is important to note that T plausibly acts in the direction opposite to the observed association – contributing towards a positive association between rehab time and injury risk. Therefore, the hazard ratio from the main Cox model (with football exposure as time axis) is plausibly biased upwards. In fact, the hazard ratio from this model is even above 1 (in contrast to the hazard ratio from the model with chronological time axis which is below 1). Considering the causal DAG (Figure 1), this could be tentatively interpreted in a way that short rehab time indicates low initial tissue defect (or highly effective rehabilitation measures) once football exposure is accounted for. However, this possibility remains to be scrutinized in future studies with more detailed medical information.

Taken together our quantitative results do not provide direct evidence that injury risk post-RTP in current elite football would be lower with longer rehab times. However, they illustrate the importance of explicitly stating causal assumptions – in addition to a specific research question – to jointly inform data analytics.

High football exposures post-RTP are observed predominantly after early return. Ancillary analyses point towards a virtual absence of high football exposures after long rehab times for the respective diagnosis (see last figure in the supplementary code file). Moreover, the combination of early return with high football exposure post-RTP seems to be a high-risk constellation. While the adjustment procedure could exaggerate football exposure in cases with a subsequent injury very early after RTP, the absence of cases with long rehab time and high football exposure impedes on estimating what would happen if rehab time is prolonged. Note that from a subject matter perspective, the effect of high football exposure post-RTP may well depend on rehab time (e.g., due to tissue healing and maturation), further questioning the validity of extrapolating beyond the range of observed values.

Come back stronger

Subtheme 1.3 ('Come back stronger.') from the qualitative part of this work, extrapolates subthemes 1.1 ('More rehab time would considerably reduce injury risk.') and 1.2 ('Biological constraints cannot be overcome.') beyond the pre-injury baseline. Note that among interviewees, opinions differed whether the potential to reduce injury risk beyond the pre-index-injury-baseline is restricted to rare cases with clearly identifiable deficits as well as about the practicability of the required timelines. While the expression 'Come back stronger.' with the above meaning is prevalent in non-scientific

communications (e.g., social media) and anecdotal evidence abounds, to the best of our knowledge the potential to use rehab time to improve on the pre-injury player state has not been systematically investigated so far. Unfortunately, the retrospective nature of our dataset did not allow for such analyses. Prospective studies in this field are therefore warranted.

Outcome oriented rehab

Beyond rehab time, suggestions for improving the rehab process (explicitly including post-RTP player management) revolved around prioritizing a 'humble', approach based on proven measures, effective communication, and a 'baby steps' progression with tight feedback loops depending on the players´ response to small increments in load (with the serious option of going a step back again). Importantly, the latter did not question the value of objective milestones at larger intervals which ensure overall structure, minimal requirements, communication and alignment of the team towards shared, verifiable goals. While guidelines and recommendations are available (Ardern et al. 2016; Keller et al. 2016), little is known about load-management during rehab in practice including the related communication and decision-making processes and finally consequences for rehab outcomes.

Smoothing out gaps and leaps

Subtheme 2.1 (Smoothing out gaps and leaps) is supported by indirect evidence from within and beyond the RTP context. Concerning avoidable 'gaps' of information and therapeutic consistency, the importance of communication quality for injury risk and injury burden in football has been documented (Ekstrand 2018; Ekstrand et al. 2019). Regarding 'leaps' in load, swift and pronounced increases in load ('spikes' (Malone et al. 2017; Windt et al. 2017)) as well as a short time period between RTP and the first match appearance (RTC) have been shown to be associated with increased injury risk (Bengtsson et al. 2020). Subtheme 2.2 (Smal steps – but no 'cowardly rehab'!) extends this continuous build-up of load and loading-capacity by calling for close-to-match loads in the final stages of rehabilitation – even if this entails the risk of negative outcomes. There is evidence that lower accumulated workloads during rehab are associated with higher risk of reinjury after RTP, e.g., for non-contact muscle injuries (Stares et al. 2018). Note that, as for the time between RTP and RTC mentioned above (Bengtsson et al. 2020), this may be due to a causal conditioning effect, including gains in trust in a controlled setting. While this is the explanation favored by the authors of the cited work (Stares et al. 2018; Bengtsson et al. 2020), a selection effect of players that would otherwise sustain a subsequent injury after RTP ('weeding out' of frail cases, cp. above) is also conceivable. Within the present work, the retrospective, media-based dataset did not allow for a specific analysis of load management during rehabilitation and the post-RTP period.

Better safe than cool

Subtheme 2.3 ('Better safe than cool!') calls for the use of proven, well understood, and well-mastered measures instead of 'the hot new stuff'. In a high-resource setting such as elite football, deliberately foregoing the newest innovations as an option for improvement seems surprising. However, it is arguably consistent with the outcome-oriented perspective of established practitioners and coincides with decision science principles, specifically the 'less can be more' effect prevailing in situations of high uncertainty (Gigerenzer et al. 2022; Hecksteden A, Keller N, et al. 2023). In other words, the imponderables of new technologies (e.g., timelines, rare or delayed adverse effects, variability of responses, etc.) are unlikely to be made up of eventual advantages over established best-practice. Note that this consideration has parallels with theme 4 ('Small can be beautiful.'): While large rehab teams are only possible in situations of abundant resources, their obvious advantages (e.g., specific expertise for all involved fields) may be negated by less obvious downsides. Examples included unclear responsibilities (and their overlap), information loss, and the need to prioritize the suitability for interpersonal communication as opposed to the ecological validity of findings (e.g., GPS data over the 'coaches' eye')).

Psychosocial factors deserve more attention

Several interviewees emphasized the need to increase the priority of psychosocial factors (theme 3, 'Squeeze and focus – Psychosocial factors deserve more attention'). Specifically, internal as well as external pressure (e.g., due to career situation or the head coach) and other psychosocial stressors may lead to pushing through minor complaints before and after RTP. This plausible, yet hard to investigate, hypothesis would compound the effect of external pressure mediated by increased football exposure post-RTP (cp. above). Also, emotional recovery and the restored ability to focus on the game are to be considered. When aiming to minimize injury risk, a psychological 'safety bubble' should be provided for the player (where possible) during rehab and the post-RTP period. While the expert stance agrees

with current guidelines advocating a biopsychosocial approach to injuries, implementation is plausibly challenging under competitive pressure.

Limitations

Interview data and thematic analysis

Interview documentation. Interviews were not recorded but documented in a written protocol. Considering the high media interest in elite football and the sensitive nature of some aspects (e.g., perceived shortcomings in a specific environment), this modification of standard proceedings was necessary to avoid restraint on the side of the interviewees and enable open communication. As it was not possible to document the interviews in verbatim throughout, quotes are limited to salient terms and phrases. Moreover, as the number of interviewees was limited, only clearly identifiable areas of overlap are reported. While this may limit the richness of qualitative results, it avoids reporting spurious findings and idiosyncratic (expert) opinion.

Injury data and quantitative analyses

Collection of injury data. There are different approaches to collecting injury data in football including collaboration with medical staff in a prospective setting (Ekstrand et al. 2021), various forms of self-report (Clarsen et al. 2014; Moller et al. 2018), media-based information and citizen-science approaches (Hoenig et al. 2022). While prospective injury registration with access to primary medical information is generally considered the reference-standard (Fuller et al. 2006), it is important to note that all approaches are associated with specific assets and drawbacks (Bjorneboe et al. 2011; Moller et al. 2018; Krutsch et al. 2020; Hoenig et al. 2022). In a field with high media coverage such as elite football, the limitations of publicly available injury data mainly concern medical information including details of injury diagnosis and the rehabilitation process. Likewise, while match exposures (minutes played) are publicly available for elite football, training exposures must be estimated based on typical values published in the literature. We made allowance for potential inaccuracies in our data by limiting the use of critical pieces of information such as index injury diagnoses.

Football exposure as time-at-risk. As illustrated in Figure 1, football exposure post-RTP is considered as the opportunity to manifest an inherent injury rate, which is itself fixed at the timepoint of RTP. This conception is reflected in the way of calculating football exposure post-RTP. However, it should be noted that football exposure might also change inherent injury risk, e.g., by inducing recovery needs. Studying the feedback loops for football exposure, player state, and injury occurrence early after RTP is clearly of interest for post-RTP player management. However, answering these additional questions requires a dynamic treatment approach (Clare et al. 2019; Shrier and Suissa 2022; Wang et al. 2024) and was beyond the scope of our study.

Global indicator of football exposure. Match and training times were integrated into a global indicator for the magnitude of opportunity to get injured ('football exposure'). As mentioned above, only match playing times were available from public sources while training times had to be estimated based on typical values published in the literature (Ekstrand et al. 2011, 2021). Considering the difference in injury incidence between training and competitive matches, simply summing up match and training times would lead to overestimating the 'opportunity to get injured' in players with low match exposure. Therefore, training times were weighted by a factor of 1/8, equally based on previous literature (Ekstrand et al. 2021; Aus der Funten et al. 2023). Such 'typical' values from external sources are associated with uncertainty. While we did not perform a sensitivity analysis (in light of the small contribution of training exposure), the potential limitation in accuracy should be kept in mind.

Distribution of episodes over the season. Football exposure varies over the competitive season, especially for players in elite clubs. While our data comprise several consecutive seasons and only episodes with full 28-day follow-up have been included, RTP from specific injuries (e.g., severe injuries) could be unevenly distributed over the season. Moreover, it seems realistic that season timepoint influences RTP decision making (and thereby rehab time) as well as football exposure post-RTP. The potential role of those interdependencies remains to be investigated in future work.

Differences between teams and leagues. Plausibly, rehabilitation and RTP environments differ, e.g., between teams and leagues. A respective random effect could account for such differences in the hierarchical Cox model. However, identifying a 'rehab environment' poses several challenges. For example, due to differences in medical care and early rehabilitation for players within the same team (specifically for severe injuries) but also due to changes between leagues or teams. Considering that expert judgment was largely consistent

between leagues and teams, 'rehab environment' has therefore not been included in the analyses.

Analysing complete episodes with full follow-up only. While the time-to-event analysis would allow for censoring by the end of the season, only complete episodes with full follow-up were included in the analyses. This decision has been made to ensure consistency with ROC curves and ancillary analyses which require a binary classification into episodes with and without a subsequent injury during the 28-day post-RTP follow-up. As the final days of the season might be different in various ways, this slightly limits the generality of our results. Moreover, cases with a very long injury-free period after RTP may be underrepresented as the respective episodes are more likely to be incomplete (that is: without observed subsequent injury) at the time point of data collection. While this consideration applies only to the last episode of a specific player, it should be kept in mind that results pertain only to players who sustained a subsequent injury.

Overall approach

Context dependence. As already discussed in more detail above, the results of this work must be considered specific to the environment in current elite European football.

Conclusion

Taken together, expert judgement (summarized in Table 1) points to options for reducing injury risk after RTP that are relatively simple – at least logistically. Most of these options for improvement are consistent with previous literature. While data analytics did not provide evidence in favour of reduced injury risk with prolonged rehabilitation, it should be kept in mind that the causal effect of rehab time is plausibly underestimated. Further research, e.g., with initial tissue defect assessed from detailed medical records, is warranted.

Moreover, supplementary analyses suggest that high football exposure occurs virtually exclusively after early return (at least in our data) and that this is a high-risk constellation. The former underlines the role of 'pressure to compete'. The latter – if replicable – entails two important messages: (1) If an early return must be attempted match playing time should be limited to the strict minimum. (2) If no injury occurred during initial match exposure(–s), luck should not be mistaken for evidence of low injury risk.

Acknowledgements

We wish to express our gratitude to all interviewees who shared their invaluable experience and insight. Without their frank and uninterested contribution, this work would not have been possible. We also wish to thank Lars Donath and Rilind Obèrtinca for helpful feedback and discussion during the drafting of this work as well as Ralph Hecksteden for expert support with data management. All authors contributed to the drafting, writing, and editing of this article. All authors have read and approved the final manuscript.

Author contributions

CRediT: **Anne Hecksteden:** Conceptualization, Data curation, Formal analysis, Funding acquisition, Investigation, Methodology, Project administration, Resources, Supervision, Writing – original draft; **Julian Berger:** Conceptualization, Data curation, Formal analysis, Investigation, Methodology, Writing – review & editing; **Raffaele Mazzolari:** Investigation, Writing – review & editing; **Guangze Zhang:** Investigation, Writing – review & editing; **Tim Meyer:** Conceptualization, Funding acquisition, Investigation, Resources, Supervision, Writing – review & editing; **Michel Brink:** Conceptualization, Data curation, Investigation, Methodology, Writing – review & editing; **Koen Lemmink:** Investigation, Supervision, Writing – review & editing; **Josef Fritz:** Formal analysis, Investigation, Methodology, Supervision, Validation, Writing – review & editing; **Niklas Keller:** Conceptualization, Investigation, Methodology, Resources, Supervision, Writing – review & editing; **Thomas Hauser:** Conceptualization, Formal analysis, Funding acquisition, Investigation, Resources, Supervision, Writing – review & editing.

Disclosure statement

Tim Meyer is the chairman of the medical committees of the German FA (DFB) and the European Football Confederation (UEFA). Anne Hecksteden and Michel Brink are associate editors of Science and Medicine in Football. The authors declare that there are no other factors that could be perceived as competing interests.

Funding

GZ is the recipient of a PhD scholarship funded by the German Football association. No other external funding was used in the preparation of this manuscript.

ORCID

Tim Meyer http://orcid.org/0000-0003-3425-4546

Data availability statement

Injury data are collected from publicly available sources. The specific data set used for the analyses presented in this work is

provided as a supplement. Interview data may not be shared due to lack of consent from the participants.

Ethics approval

The study was approved by the institutional review board of the Institute of Sport Science at the University of Innsbruck (approval number 57/2023 and amendment).

Informed consent

All interviewees were informed about the rationale and proceeding of the study and provided informed consent prior to participation.

References

Ardern CL et al. 2016. 2016 consensus statement on return to sport from the first World Congress in Sports Physical Therapy, Bern. Br J Sports Med. 50(14):853–864. https://doi.org/10.1136/bjsports-2016-096278

Arnason A et al. 2004. Risk factors for injuries in football. Am J Sports Med. 32(1 Suppl):5S–16S. https://doi.org/10.1177/0363546503258912

Aus der Funten K et al. 2023. Epidemiology of football injuries of the German Bundesliga: a media-based, prospective analysis over 7 consecutive seasons. Sports Med Open. 9(1):20. https://doi.org/10.1186/s40798-023-00563-x

Bahr R et al. 2020. International Olympic Committee consensus statement: methods for recording and reporting of epidemiological data on injury and illness in sport 2020 (including STROBE extension for sport injury and illness surveillance (STROBE-SIIS)). Br J Sports Med. 54(7):372–389. https://doi.org/10.1136/bjsports-2019-101969

Barnes C, Archer DT, Hogg B, Bush M, Bradley PS. 2014. The evolution of physical and technical performance parameters in the English Premier League. Int J Sports Med. 35(13):1095–1100. https://doi.org/10.1055/s-0034-1375695

Beischer S et al. 2020. Young athletes who return to sport before 9 months after anterior cruciate ligament reconstruction have a rate of new injury 7 times that of those who delay return. J Orthop Sports Phys Ther. 50(2):83–90. https://doi.org/10.2519/jospt.2020.9071

Bengtsson H, Ekstrand J, Walden M, Hagglund M. 2020. Few training sessions between return to play and first match appearance are associated with an increased propensity for injury: a prospective cohort study of male professional football players during 16 consecutive seasons. Br J Sports Med. 54(7):427–432. https://doi.org/10.1136/bjsports-2019-100655

Benjaminse A, Verhagen E. 2021. Implementing ACL injury prevention in daily sports practice-it's not just the program: let's build together, involve the context, and improve the content. Sports Med. 51(12):2461–2467. https://doi.org/10.1007/s40279-021-01560-4

Bittencourt NFN et al. 2016. Complex systems approach for sports injuries: moving from risk factor identification to injury pattern recognition-narrative review and new concept. Br J Sports Med. 50(21):1309–1314. https://doi.org/10.1136/bjsports-2015-095850

Bjorneboe J, Florenes TW, Bahr R, Andersen TE. 2011. Injury surveillance in male professional football; is medical staff reporting complete and accurate? Scand J Med Sci Sports. 21(5):713–720. https://doi.org/10.1111/j.1600-0838.2009.01085.x

Braun V, Clarke V, Weate P. 2016. Using thematic analysis in sport and exercise research. In: Smith B, Sparkes A, editors. Routledge handbook of qualitative research in sport and exercise. Routledge.

Clare PJ, Dobbins TA, Mattick RP. 2019. Causal models adjusting for time-varying confounding-a systematic review of the literature. Int J Epidemiol. 48(1):254–265. https://doi.org/10.1093/ije/dyy218

Clarsen B, Ronsen O, Myklebust G, Florenes TW, Bahr R. 2014. The Oslo Sports Trauma Research Center Questionnaire on Health Problems: a new approach to prospective monitoring of illness and injury in elite athletes. Br J Sports Med. 48(9):754–760. https://doi.org/10.1136/bjsports-2012-092087

Della Villa F, Hagglund M, Della Villa S, Ekstrand J, Walden M. 2021. High rate of second ACL injury following ACL reconstruction in male professional footballers: an updated longitudinal analysis from 118 players in the UEFA elite club injury study. Br J Sports Med. 55(23):1350–1356. https://doi.org/10.1136/bjsports-2020-103555

Edouard P, Caumeil B, Verhagen E, Guilhem G, Ruffault A. 2022. Maximising individualisation of sports injury risk reduction approach to reach success. Braz J Phys Ther. 26(3):100394. https://doi.org/10.1016/j.bjpt.2022.100394

Ekstrand J. 2018. Is there a correlation between coaches' leadership styles and injuries in elite football teams? A study of 36 elite teams in 17 countries. Br J Sports Med. 52(8):527–531. https://doi.org/10.1136/bjsports-2017-098001

Ekstrand J, Hagglund M, Walden M. 2011. Injury incidence and injury patterns in professional football: the UEFA injury study. Br J Sports Med. 45(7):553–558. https://doi.org/10.1136/bjsm.2009.060582

Ekstrand J, Lundqvist D, Davison M, D'Hooghe M, Pensgaard AM. 2019. Communication quality between the medical team and the head coach/manager is associated with injury burden and player availability in elite football clubs. Br J Sports Med. 53(5):304–308. https://doi.org/10.1136/bjsports-2018-099411

Ekstrand J, Spreco A, Bengtsson H, Bahr R. 2021. Injury rates decreased in men's professional football: an 18-year prospective cohort study of almost 12 000 injuries sustained during 1.8 million hours of play. Br J Sports Med. 55(19):1084–1091. https://doi.org/10.1136/bjsports-2020-103159

Fuller CW et al. 2006. Consensus statement on injury definitions and data collection procedures in studies of football (soccer) injuries. Br J Sports Med. 40(3):193–201. https://doi.org/10.1136/bjsm.2005.025270

Gajhede-Knudsen M, Ekstrand J, Magnusson H, Maffulli N. 2013. Recurrence of achilles tendon injuries in elite male football players is more common after early return to play: an 11-year follow-up of the UEFA Champions League injury study. Br J Sports Med. 47(12):763–768. https://doi.org/10.1136/bjsports-2013-092271

Gigerenzer G, Reb J, Luan S. 2022. Smart heuristics for individuals, teams, and organizations. Annu Rev Organ Psychol Organ Behav. 9(1):171–198. https://doi.org/10.1146/annurev-orgpsych-012420-090506

Greenland S, Mansournia MA, Altman DG. 2016. Sparse data bias: a problem hiding in plain sight. BMJ. i1981. https://doi.org/10.1136/bmj,i1981

Hagglund M et al. 2013. Injuries affect team performance negatively in professional football: an 11-year follow-up of the UEFA Champions League injury study. Br J Sports Med. 47 (12):738–742. https://doi.org/10.1136/bjsports-2013-092215

Hagglund M, Walden M, Ekstrand J. 2007. Lower reinjury rate with a coach-controlled rehabilitation program in amateur male soccer: a randomized controlled trial. Am J Sports Med. 35(9):1433–1442. https://doi.org/10.1177/0363546507300063

Hagglund M, Walden M, Ekstrand J. 2016. Injury recurrence is lower at the highest professional football level than at national and amateur levels: does sports medicine and sports physiotherapy deliver? Br J Sports Med. 50 (12):751–758. https://doi.org/10.1136/bjsports-2015-095951

Hecksteden A et al. 2023. Forecasting football injuries by combining screening, monitoring and machine learning. Sci Med Footb. 7(3):214–228. https://doi.org/10.1080/24733938.2022.2095006

Hecksteden A, Keller N, Zhang G, Meyer T, Hauser T. 2023. Why humble farmers may in fact grow bigger potatoes: a call for street-smart decision-making in sport. Sports Med - Open. 9 (1):94. https://doi.org/10.1186/s40798-023-00641-0

Hecksteden A, Kellner R, Donath L. 2022. Dealing with small samples in football research. Sci Med Football. 6(3):389–397. https://doi.org/10.1080/24733938.2021.1978106

Hoenig T et al. 2022. Analysis of more than 20,000 injuries in European professional football by using a citizen science-based approach: an opportunity for epidemiological research? J Sci Med Sport. 25(4):300–305. https://doi.org/10.1016/j.jsams.2021.11.038

Katsikopoulos K, Ö S, Buckmann M, Gigerenzer G. 2021. Classification in the wild: the science and art of transparent decision making. The MIT press.

Keller M, Kurz E, Schmidtlein O, Welsch G, Anders C. 2016. Interdisziplinäre Beurteilungskriterien für die Rehabilitation nach Verletzungen an der unteren Extremität: Ein funktionsbasierter Return-to-Activity Algorithmus. Phys Med Rehab Kuror. 26(3):137–148. https://doi.org/10.1055/s-0042-106599

Krutsch V et al. 2020. Injury analysis in professional soccer by means of media reports - only severe injury types show high validity. Open Access J Sports Med. 11:123–131. https://doi.org/10.2147/OAJSM.S251081

Lutter C, Jacquet C, Verhagen E, Seil R, Tischer T. 2022. Does prevention pay off? Economic aspects of sports injury prevention: a systematic review. Br J Sports Med. 56(8):470–476. https://doi.org/10.1136/bjsports-2021-104241

Malone S, Roe M, Doran DA, Gabbett TJ, Collins KD. 2017. Protection against spikes in workload with aerobic fitness and playing experience: the role of the acute: chronic workload ratio on injury risk in elite Gaelic football. Int J Sports Physiol Perform. 12(3):393–401. https://doi.org/10.1123/ijspp.2016-0090

Moller M et al. 2018. Validity of the SMS, phone, and medical staff examination sports injury surveillance system for time-loss and medical attention injuries in sports. Scand J Med Sci Sports. 28(1):252–259. https://doi.org/10.1111/sms.12869

Obertinca R et al. 2024. Efficacy of a new injury prevention programme (FUNBALL) in young male football (soccer) players: a cluster-randomised controlled trial. Br J Sports Med. 58 (10):548–555. https://doi.org/10.1136/bjsports-2023-107388

O'Brien BC, Harris IB, Beckman TJ, Reed DA, Cook DA. 2014. Standards for reporting qualitative research: a synthesis of recommendations. Acad Med. 89(9):1245–1251. https://doi.org/10.1097/ACM.0000000000000388

Orchard J, Best TM. 2002. The management of muscle strain injuries: an early return versus the risk of recurrence. Clin J Sport Med. 12(1):3–5. https://doi.org/10.1097/00042752-200201000-00004

Ruddy JD et al. 2019. Modeling the risk of team sport injuries: a narrative review of different statistical approaches. Front Physiol. 10:829. https://doi.org/10.3389/fphys.2019.00829

Rudin C. 2019. Stop explaining black box machine learning models for high stakes decisions and use interpretable models instead. Nat Mach Intell. 1(5):206–215. https://doi.org/10.1038/s42256-019-0048-x

Rudin C et al. 2021. Interpretable machine learning: fundamental principles and 10 grand challenges. Preprint. https://arxivorg/abs/210311251v2.2021

Shrier I. 2015. Strategic assessment of risk and risk tolerance (StARRT) framework for return-to-play decision-making. Br J Sports Med. 49(20):1311–1315. https://doi.org/10.1136/bjsports-2014-094569

Shrier I, Suissa S. 2022. The quintessence of causal DAGs for immortal time bias: time-dependent models. Int J Epidemiol. 51(3):1028–1029. https://doi.org/10.1093/ije/dyac056

Stares J et al. 2018. How much is enough in rehabilitation? High running workloads following lower limb muscle injury delay return to play but protect against subsequent injury. J Sci Med Sport. 21(10):1019–1024. https://doi.org/10.1016/j.jsams.2018.03.012

Tabben M et al. 2023. Obstacles and opportunities for injury prevention in professional football in Qatar: exploring the implementation reality. BMJ Open Sport Exerc Med. 9(1): e001370. https://doi.org/10.1136/bmjsem-2022-001370

Timans R, Wouters P, Heilbron J. 2019. Mixed methods research: what it is and what it could be. Theor Soc. 48 (2):193–216. https://doi.org/10.1007/s11186-019-09345-5

van de Schoot R et al. 2021. Bayesian statistics and modelling. Nat Rev Mater. 1(1). https://doi.org/10.1030/s43586-020-00001-2

Wang C, Kaufman JS, Steele RJ, Shrier I. 2024. Target trial framework for determining the effect of changes in training load on injury risk using observational data: a methodological commentary. BMJ Open Sport Exerc Med. 10(3):e002037. https://doi.org/10.1136/bmjsem-2024-002037

Wangensteen A et al. 2016. Hamstring reinjuries occur at the same location and early after return to sport: a descriptive study of MRI-confirmed reinjuries. Am J Sports Med. 44 (8):2112–2121. https://doi.org/10.1177/0363546516646086

Windt J, Zumbo BD, Sporer B, MacDonald K, Gabbett TJ. 2017. Why do workload spikes cause injuries, and which athletes

are at higher risk? Mediators and moderators in workload-injury investigations. Br J Sports Med. 51 (13):993–994. https://doi.org/10.1136/bjsports-2016-097255

Yung KK, Ardern CL, Serpiello FR, Robertson S. 2022. A framework for clinicians to improve the decision-making process in return to sport. Sports Med Open. 8(1):52. https://doi.org/10.1186/s40798-022-00440-z

Zhang G et al. 2025. The time course of injury risk after return-to-play in professional football (soccer). Sports Med. 55(1):193–201. https://doi.org/10.1007/s40279-024-02103-3